Thoughts on Mathematical Values

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Abstract. *The development of AI may change what is possible, but it does not have to determine what we choose to value. I hope this essay encourages a philosophical reflection on these questions.*

One might reasonably think that if AI could produce genuinely good mathematics, we could appreciate it provided that mathematicians are willing to understand the results. It is also quite possible that conceptual development and mathematical judgment would remain particularly difficult for AI even though AI became good at some kinds of problem solving, so that the mathematical community might naturally shift its values towards aspects where AI is less capable.

I still find these considerations reasonable. But they also raise a philosophical question: How much should our mathematical values depend on what AI is capable of? The answer may be quite personal and may also vary across different fields of mathematics.

What Makes Mathematics Valuable?

There are many different mathematical theories and research directions, all of which can be good mathematics. But mathematicians do not value them equally. Different fields have their own traditions, tastes, and standards of what counts an interesting contribution. These judgments are shaped not only by mathematical difficulty or applications, but also by history, individual personalities, and intellectual traditions.

There is also considerable diversity in what different areas of mathematics value. Some mathematical traditions place greater emphasis on obtaining concrete results, solving problems, or improving existing results, but others emphasize more on conceptual understanding, aesthetics, and the craftsmanship of mathematical discovery. These distinctions are certainly not absolute, but they may help explain why mathematicians have very different attitudes toward AI.

What I find particularly interesting is that these attitudes are not determined solely by what AI is capable of doing. If a community places great value on independent human discovery, conceptual developments and its intellectual traditions, its members may naturally be less inclined to rely on AI, even when it could produce useful results. Conversely, if AI-assisted results are increasingly valued, their use may gradually become a normal part of mathematical practice in some communities. Different mathematical communities have considerable power to shape their own values, which in turn influence the attitudes of their individual members. What mathematicians choose to admire, encourage, and pass down to younger generations helps shape the traditions of how mathematics is approached and pursued.

In this sense, the future of mathematics is not simply something determined by technological developments. It is also shaped by the collective choices of mathematicians themselves. Different mathematical traditions may reasonably choose different paths, and I think we should recognize this freedom rather than assuming that increasing AI use is inevitable or desirable everywhere.

But maintaining this freedom and to have control requires collective effort. For example, mathematical communities have traditionally placed considerable value on being the first to obtain a result. But when the subjects producing such results are replaced by AI, we need

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not apply the same standards of recognition. An independent human proof may deserve the recognition traditionally associated with a first proof, even if an AI-generated proof already exists. More fundamentally, the importance attached to being first is itself worth reconsidering. Mathematics is not simply a competition to solve problems first. Different methods and ways of understanding the same result can be equally important whether or not they are the first.

Of course, there is a reasonable attitude that good mathematics should be appreciated regardless of how it was produced. I still sympathize with this view. But the mathematical community has considerable freedom to decide what counts as good mathematics and what kinds of contributions it values. This distinction becomes especially important now since valuing "good" mathematics is no longer necessarily the same as valuing human mathematical activity or understanding.

Mathematics and Artistic Independence

Artists and musicians often have strong ideas about what makes their work valuable. They may choose to pursue their artistic ideals instead of following commercial trends, despite the uncertainty this implies. A similar attitude can be found in traditional craftsmanship. Although machines can produce musical instruments with remarkable quality and precision, the work of luthiers remains deeply valued for the individual characters and personalities of the sound in their instruments.

Could mathematics develop along similarly different paths? Perhaps some mathematical communities will increasingly embrace AI as a means of producing results, while others will continue to cherish the craftsmanship of mathematical discovery, preserving traditions of understanding and individual insight. After all, mathematics is not just about producing results, just as making musical instruments is not just about producing sound.

I am not sure how much these differences arise from the nature of the mathematics itself and how much from the intellectual cultures of individual fields. Perhaps AI will amplify differences in mathematical values that already exist across different mathematical communities.

Maintaining intellectual independence can be difficult especially for people facing career uncertainty and competitive pressures. Some worry that AI will make academic competition even more unfair, but I am not convinced that the evaluation system is necessarily so vulnerable. Uncertainty existed long before AI. Academic careers have never depended entirely on mathematical ability or the quality of one's work. Advisors' recommendations, connections, research fashions, and luck have always mattered the most. Moreover, academic hiring considers more than publications, it already includes qualities that cannot simply be replaced by AI-generated results. AI does not necessarily make the situation worse.

Of course, we should not ignore new concerns and difficulties arising from AI. There is no universal solution to these difficulties. But perhaps part of intellectual independence is accepting that doing what we value does not always bring the outcomes we hope for. This uncertainty is part of the risk of intellectual freedom.

Developing Mathematical Understanding

AI can still be very useful in many ways, like exploring the literature, finding connections and getting general overviews. These uses may not be so different from searching the internet or consulting the literature and AI can make such exploration much more convenient. But when it comes to mathematical arguments themselves, I think we should be more cautious about the role we allow AI to play. I do not think we should ever adopt a research style in which

humans provide only conceptual judgment while leaving the mathematical details to AI. At the very least, authors should be able to check and understand all the details themselves. But understanding is itself difficult to judge. Knowing how much we really understand is a skill that develops with experience. Sometimes it takes being repeatedly challenged or reminded that we have not understood something to develop such judgment.

I remember fondly how my commutative algebra mentor could often look at a complicated proof and say that there was a much simpler argument once one understood what was really going on. I also learned from working with different professors that there are different ways of engaging with details to develop understanding. The details of a proof can also reveal a lot about how a mathematician understands mathematics.

I also feel lucky to have inherited a distinctive mathematical perspective from my algebra mentor, which has given me directions to explore for years to come. I suppose such insights come from deep understanding and also have something to do with individual mathematical personalities. They develop through sustained engagement with mathematical details, including work that may not appear directly useful. Like a plant, it takes time and nourishment to grow. Trying to accelerate the process too much risks destroying what we cherish.



Nurturing Mathematical Insight

The boundary between human and AI involvement in mathematical research is not just about technological capabilities but also of what we value in mathematics. Our caution toward encouraging AI comes partly from our belief in the value of mathematical understanding and intellectual craftsmanship, and partly from our attachment to an intellectual tradition that we deeply cherish.

At the same time, we generally trust mathematicians who have developed mature mathematical judgment to decide how best to use these tools. We are not suggesting that most mathematicians use AI without understanding, and we recognize that many ways of using AI in research are perfectly respectable. We also believe that the use of AI does not necessarily diminish the intellectual engagement and that where one chooses to draw the boundary may ultimately be a matter of personal values.

Continuing to Enjoy Mathematics

Perhaps the most important thing is that mathematics should remain something we genuinely enjoy doing.

Mathematical values need not remain unchanged forever. But changes in what we value should come from our understanding of mathematics and our intellectual judgment, rather than simply from technological capabilities or the practices of others.

Perhaps mathematicians can be a little stubborn about the things they find valuable. The point is not simply to produce as much mathematics as possible, or to compete with whatever becomes capable of producing it. It is also to continue finding mathematics worth thinking about, understanding, and enjoying.



Mathematical Stubbornness

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